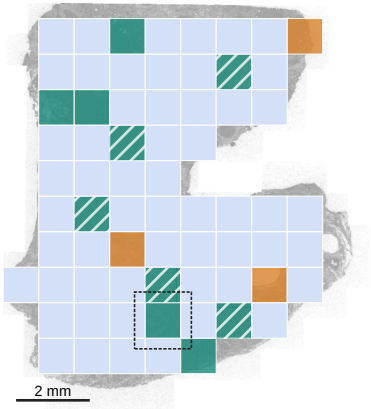
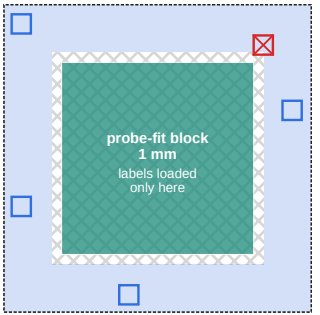
a Split within a section: 1 mm blocks, fixed once



zoom (dashed box in left map)



hatched: 50 µm guard band, molecules dropped; 100 µm training windows (blue) may not reach it (red: rejected)

train 80% · validation 5%

probe-fit 7.5% · probe-test 7.5%

b Held-out ladder: every claim names its rung

harder generalization

4	**a whole platform** — platform shift	phase 4
3	**dataset with a new panel** — panel shift	phase 4
2	**all sections of a donor** — donor variation	phase 3, 5
1	**whole sections, seen donor** — section variation	phase 3
0	**1 mm blocks** — fit, not generalization	phases 1–2

c Phases: each ends at a gate; a failed gate stops the next phase

	Phase	Gate
0	**Data and vocabulary** HGNC vocabulary gene prior emb. split file, baselines	every gene → one token; split frozen; baselines scored
1	**Single-panel pilot** 1 Xenium section CellWorld-patch falsification first	no collapse; beats composition at 2 of 3 scales
2	**Panel robustness** panel dropout thinning, count noise panel/platform tokens	≥80% retention on 35%-overlap split; <5% IID cost
3	**Multi-section** 10–20 sections 2–3 platforms random context size	beats composition on rungs 1–2
4	**Cross-platform transfer** zero-shot, then platform token, then last block or LoRA	beats composition on shared genes, cell and niche
5	**Scale up** more donors model size; data fractions by section	final numbers on held-out tissue (rungs 2–4)

recipe throughout: AdamW, BF16, batch 256, warmup to 3e-3 then cosine, EMA 0.999 → 0.9999, 10% hint, 10k steps in the pilot; block masking at 5/10/50 µm, random masking as first ablation; adaptation only via platform token, last block or LoRA.